

Assignment 4: Optimizing Transformer Translation with Ray Tune & Optuna

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1 Introduction

This report describes the optimization of a custom Transformer model for English-to-Hindi translation using **Ray Tune** with the **Optuna** search algorithm and the **ASHA** (Asynchronous Successive Halving Algorithm) scheduler. The objective is to match or exceed the BLEU score achieved by the baseline model (trained for 100 epochs) in significantly fewer epochs, thereby reducing wall-clock training time by a substantial margin.

Links

- **GitHub Repository:** <https://github.com/Laksh-Mendpara/MLOps-Laksh-B23CS1037/tree/Assignment-4>
- **Hugging Face Model:** <https://huggingface.co/Laksh-Mendpara/MLOPS-Assignment-4/tree/main>

2 Baseline Metrics

The baseline model is a standard Transformer (6 encoder / 6 decoder layers, $d_{\text{model}} = 512$, 8 heads, $d_{\text{ff}} = 2048$, dropout= 0.1) trained on the English–Hindi Tatoeba corpus ($\sim 13\,186$ sentence pairs) for **100 epochs** with a batch size of 60 and a learning rate of 10^{-4} .

Table 1: Baseline performance (100 epochs).

Metric	Value
Number of Epochs	100
Training Time	~ 70 min
Final Training Loss	0.0976
BLEU Score (NLTK)	75.66

The BLEU score was evaluated on a small held-out set of 5 sentence pairs using NLTK’s `corpus_bleu` with `SmoothingFunction` method4.

3 Hyperparameter Search Space

We tuned **five** hyperparameters using the Optuna search algorithm. The search space is summarised in Table 2.

Notes:

Table 2: Hyperparameter search space.

Hyperparameter	Type	Range / Choices	Scale
Learning Rate (<code>lr</code>)	Continuous	$[1 \times 10^{-5}, 1 \times 10^{-3}]$	Log-uniform
Batch Size (<code>batch_size</code>)	Categorical	$\{16, 32, 64\}$	—
Attention Heads (<code>num_heads</code>)	Categorical	$\{4, 8\}$	—
Feed-Forward Dim (<code>d_ff</code>)	Categorical	$\{1024, 2048\}$	—
Dropout (<code>dropout</code>)	Continuous	$[0.1, 0.4]$	Uniform

- The model dimension $d_{\text{model}} = 512$ and number of encoder/decoder layers (6 each) were kept fixed.
- Each trial was allowed up to **20 epochs**; the ASHA scheduler aggressively pruned under-performing trials using a grace period of 3 epochs with a reduction factor of 3.
- We ran **20 trials** concurrently across 4 GPUs (one trial per GPU) to maximize hardware utilization.

4 Best Configuration

The best configuration identified by the Optuna-guided search is presented in Table 4.

Table 3: Best hyperparameter configuration found.

Hyperparameter	Optimal Value
Learning Rate (<code>lr</code>)	1.4925×10^{-4}
Batch Size (<code>batch_size</code>)	16
Attention Heads (<code>num_heads</code>)	4
Feed-Forward Dim (<code>d_ff</code>)	1024
Dropout (<code>dropout</code>)	0.335

Key observations:

- The optimal learning rate ($\sim 1.5 \times 10^{-4}$) is very close to the baseline value of 10^{-4} , but slightly higher, enabling faster convergence within the 20-epoch budget.
- A *smaller* batch size (16 vs. 60 in the baseline) introduces more gradient noise, which acts as an implicit regulariser and aids generalisation.
- Reducing the number of heads from 8 to 4 and halving d_{ff} from 2048 to 1024 simplifies the model, reducing overfitting on the relatively small ($\sim 13\text{k}$ pairs) dataset while retaining sufficient capacity.
- The higher dropout (0.335 vs. 0.1) further regularises the model and complements the smaller batch size.

5 Final Optimized Model Metrics

After identifying the best configuration, the model was retrained from scratch for 20 epochs. The final metrics are compared with the baseline in Table 4.

Summary of improvements:

- **BLEU score improved** from 75.66 to **90.38** (+14.72 points), significantly surpassing the baseline.
- **Training epochs reduced** from 100 to **20** ($5\times$ reduction).
- **Wall-clock time reduced** from ~ 70 min to ~ 24 min ($\sim 2.9\times$ speed-up).

Table 4: Baseline vs. Optimized model comparison.

Metric	Baseline	Optimized
Number of Epochs	100	20
Training Time	~70 min	~ 24 min
Final Training Loss	0.0976	0.8760
BLEU Score (NLTK)	75.66	90.38

- Although the training loss is higher (0.876 vs. 0.098), the substantially higher BLEU score indicates better generalisation. The baseline likely overfits on the small training corpus over 100 epochs, memorising training tokens rather than learning transferable translation patterns.

6 Conclusion

By leveraging Ray Tune with Optuna and the ASHA scheduler, we were able to efficiently explore the hyperparameter space and identify a configuration that *significantly outperforms* the baseline in translation quality (BLEU 90.38 vs. 75.66) while requiring only 20% of the training epochs and ~34% of the wall-clock time. The combination of a smaller batch size, moderate dropout, and a leaner feed-forward network proved most effective for this relatively small parallel corpus.